SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07 02

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

20/20

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

I SHAIK KHAJAM ROSHAN 192511445 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Q.1 Star Topology Vs Bus Topology

Star Topology:

- It connects every device to a central device such as a switch or hub.

Bus Topology:

- It connects all devices through a single backbone cable.

Cost:

- Bus topology is less expensive because it only requires one backbone cable and fewer connections.
- Star topology requires separate cables from every device to central switch / hub, increase intallation cost.

Reliability:

- Star topology generally provides better performance because switch manages communication between devices.

• Bus topology, failure of main backbone cable can affect entire network.

Performance:

• Star topology generally provides better than bus topology.

Example:

• Star topology is for office network.
• Bus topology is for small network.

Conclusion:

• Bus is cheaper and simple but star provides better reliability and performance.

Q.2 Mesh Topology Vs Star Topology

Mesh Topology:

• It provides multiple connections between nodes.

Star Topology:

• It connects all devices through a central switch or hub.

Fault Tolerance

• Mesh topology has very high fault tolerance because multiple paths are available.

- Star topology has moderate fault tolerance because of central hub or switch.

Reliability:

- Mesh is more reliable because communication can continue through alternate paths. Star depends heavily on its central device.

Analytical Comparison:

- For a network where continuous communication is critical, Mesh is preferable.
- Star is more suitable when simple install and management are required.

Example:

- Mesh topology can be used in critical communication enenvironments.
- Star topology is used where network available is very important.

Conclusion:

- Mesh topology is more fault-tolerant
- Star topology is more reliable

Q.3 Bus Topology Vs Mesh Topology

Bus Topology:

- It is simple and easy to install because it requires few cables and connections.

Mesh Topology:

- It is difficult to install becaush each node requires multiple connections.

Installation:

- Bus Topology is simple and easy to install because it needs fewer cables.
- Mesh Topology is difficult and difficult to install because of multiple nodes.

Complexity:

- The complexity of Mesh increases significatly when more devices added.
- This results is more cables, ports and connections is managed.

Cost:

- Bus topology is cheapher because less cables and few networking.

• Mesh is more hardware and cabling making it more expensive.

Scalability

• In Mesh increases installation complexity.
• In Bus shared backbone limits its for large networks.

Example:

• Bus topology with limited devices and low cost

Conclusion:

• Bus topology is cheaper and less complex.
• Mesh topology is expensive and more complex.

Q.4 Hybrid Topology Vs Star Topology

Hybrid Topology:

• It combines 2 or more different topologies such as Star, Mesh or Bus.

Star Topology:

• It connects devices through central switch or hub.

Scalability:

• Hybrid topology provides greater scalability because different network sections.

- Star topology is also scalable because of different available ports in centrale hub or switch.

Flexibility:

- Hybrid topology is more flexible because of different topologies can be combined and based on requirements

Analytical Comparison:

- For a small network "Star is sufficient"
- For large organization "Hybrid provides best flexibility".

Example:

- Star topology is for one department.
- Hybrid topology is for one organization.

Conclusion:

- Hybrid topology provides greater scalability and flexiblity than star topology.

Q.5 Maintenance Comparison of Network Topologies

Star Topology:

- It is harder to troubleshoot because a problem in backbone cable can affect complete network

Mesh Topology:

- It is highly reliable but difficult to maintain large number of connections.

Bus Topology:

- When a problem occurs, administrators can identify & isolate affected node.

Hybrid Topology:

- It is moderate maintenance complexity and combines different toopology.

Analytical Comparison

- "Star" is easier to maintain.
- "Mesh" requires more effort to connect.
- "Bus" can be difficult when backbone is failed.
- "Hybrid" requires skilled management.

Example:

- Star topology is suitable for organizations when quick fault identification
- Easy maintenance are important.

Conclusion:

- Star topology is generally the easiest to maintain and handle.
- Mesh is more complex to trobleshoot and mange.
- Hybrid is high scalable and flexible.